



Nazmi Yücel Çan

MACHINE LEARNING ENGINEER • Giresun, Türkiye (Open to Relocation / Remote)

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TECHNICAL PROFILE

Machine Learning Engineer specializing in model development, optimization, and edge AI deployment. Reduced mT5 SLM model size 5x (from 500MB to 100MB) via pruning, 8-bit LoRA, and INT8 quantization. Following computer vision experience, currently focusing on Natural Language Processing (NLP). Co-authored a research paper published in El-Cezeri Journal (2026). Developed Zettelkasten AI Notes (Pupilica AI Hackathon Top 10 Finalist), featuring a Two-Tier Vector RAG pool, CPU cross-encoder reranking, and Vulkan GPU local GGUF inference.

SKILLS

Programming Languages

Python (Advanced), SQL, Bash/Shell Scripting

ML & AI Frameworks

TensorFlow, PyTorch, Keras, Scikit-learn,
Hugging Face (LoRA/PEFT), ONNX Runtime,
llama-cpp-python

Core Domains

Computer Vision (OpenCV, YOLOv8), NLP
(Transformers, RAG & Embeddings), Data
Preprocessing & Evaluation, Model Optimization
(Quantization & Pruning), MLOps Fundamentals

Tools & Platforms

Git/GitHub, Linux (Ubuntu/Debian), PyQt5, Flet, Google Gemini API

Languages: Turkish (Native), English (Upper-Intermediate)

EXPERIENCE

Machine Learning Intern at Deep Think Technology

02/2026 – 06/2026

- Developed the industrial "Apple Detection and Quality Analysis System", integrating real-time YOLOv8-seg fruit detection, CIEDE2000 color quality analysis, physical dimension measurement, and camera SDK into a unified PyQt5 desktop application.
- Trained and deployed computer vision models on embedded devices for real-time edge AI inference.
- Implemented data preprocessing and model evaluation workflows for edge computing hardware, addressing real-world latency and resource constraints.

PROJECTS

mT5 Bidirectional EN-TR Machine Translation (ONNX INT8 & ORT Mobile)

Python, PyTorch, Transformers, LoRA, Fine-Tuning, ONNX Runtime, NLP

- Fine-tuned an mT5-small model with 8-bit LoRA for bidirectional English-Turkish translation, achieving a 0.6245 METEOR score. Reduced model size 5x (from 500MB to 100MB) and optimized performance for resource-constrained environments by applying model pruning, ONNX compilation, and INT8 quantization for mobile deployment via ONNX Runtime (ORT Mobile).

Kaggle

Zettelkasten AI Notes (Pupilica AI Hackathon — Top 10 Finalist)

Python, Flet, SQLite (WAL), RAG, llama.cpp (Vulkan), Cross-Encoder, Gemini API, NLP

- A privacy-first desktop knowledge management application implementing the Zettelkasten method using Python and Flet. Features a Dual AI Engine (Google Gemini and Vulkan-accelerated local GGUF models) with a Two-Tier In-Memory Vector RAG Pool, CPU cross-encoder reranking (Qwen3), and local embeddings (Harrier-0.6B) to synthesize atomic notes without duplicate concepts. Includes an interactive canvas mind map, live Markdown editor with LaTeX, smart WikiLinks, and SQLite WAL persistence.

GitHub

Apple Detection and Quality Analysis System

Python, OpenCV, YOLO, PyQt5, Computer Vision

- An industrial computer vision application featuring real-time object detection, physical dimension measurement, and CIEDE2000 color quality analysis. Focused on system integration, designing the application architecture, and orchestrating the end-to-end pipeline—integrating the camera SDK, real-time YOLOv8-seg model inference, and measurement modules into a unified desktop interface.

[GitHub](#)

Arduino PC System Monitor

Python, Arduino, TinyML, scikit-learn

- A TinyML-powered system that monitors PC hardware resources (CPU, RAM, GPU, VRAM) and automatically optimizes power consumption. The system extracts hardware telemetry in real-time, executing an on-device SVM classification model on an Arduino Leonardo to determine and adjust the system's power state.

[GitHub](#)

PUBLICATIONS

Deep Learning Models Integrating Attention Mechanisms For Military Camouflaged Object Detection — Nilgün Şengöz, Gül Karaman, Mert Samet Çeliker, Nazmi Yücel Çan

El-Cezeri Fen ve Mühendislik Dergisi (2026) — <https://doi.org/10.31202/ecjse.1747013>

EDUCATION

Bachelor of Science in Information Systems and Technologies

09/2022 – 06/2026

Burdur Mehmet Akif Ersoy University

GPA: 3.17/4.00

REFERENCES

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